

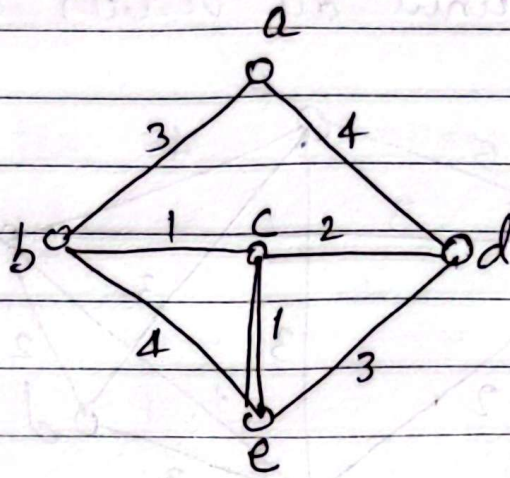
Algorithms For minimal Spanning Tree

1. Kruskal's Algorithm.

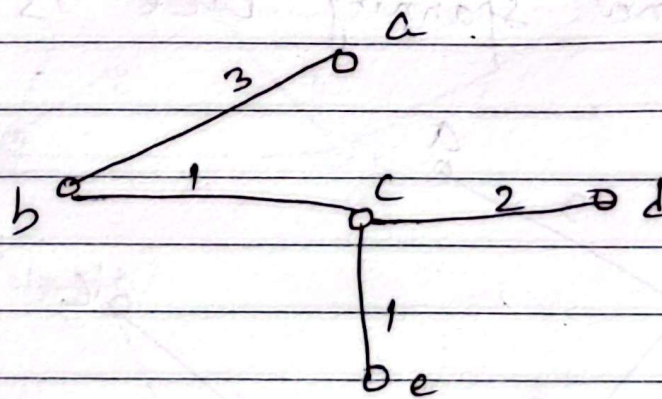
Working Rule:

- (i) Choose an edge of minimal weight.
- (ii) At each step, choose the edge whose inclusion will not create a circuit.
- (iii) If G has n vertices, stop after $(n-1)$ edges.

Example:



Its spanning tree is



∴ Total wt. of ^{minimum} spanning tree = $3+1+1+2 = 7$.

Note: The weight of a minimal spanning tree is unique, but the minimal spanning tree itself is not. Different minimal spanning trees can occur when two or more edges have the same weight.

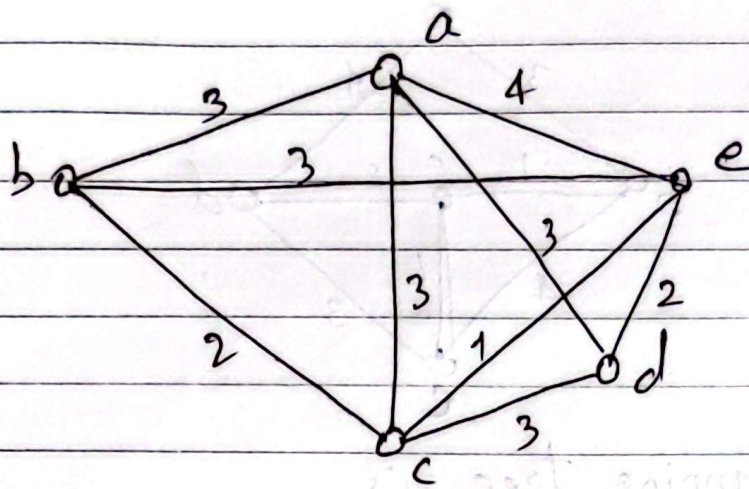
Notes

2. Prim's Algorithm

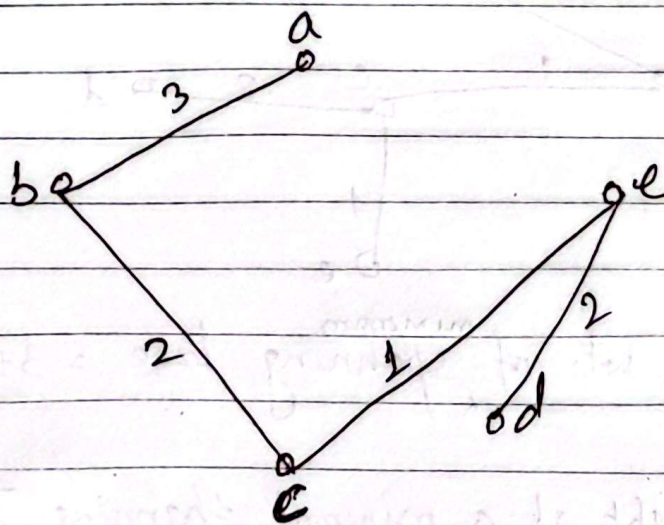
Working Rule:

- (i) Select any vertex and choose the edge and smallest weight from G .
- (ii) At each stage, choose the edge of smallest weight. Joining a vertex already included to vertex not yet included.
- (iii) Continue until all vertices are included.

e.g.

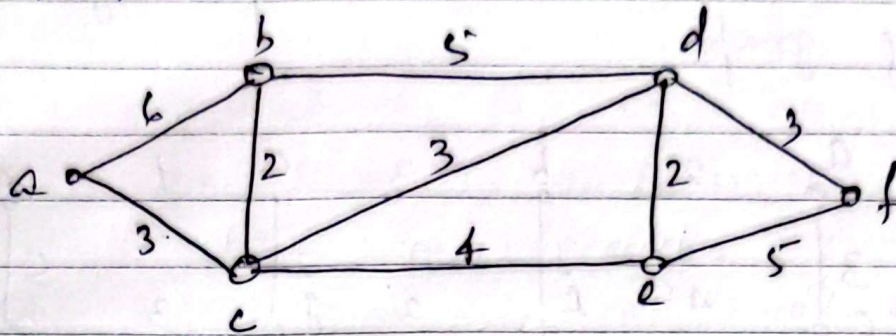


Its minimal spanning tree is



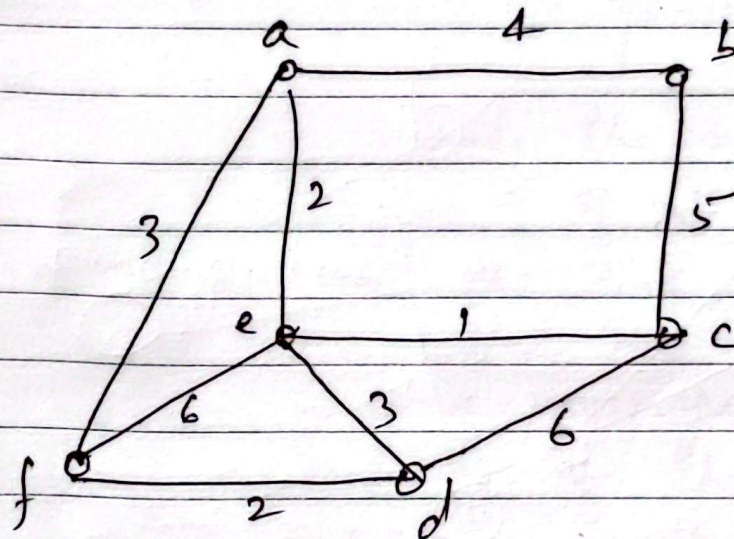
$$\text{Total wt.} = 3 + 2 + 1 + 2 = 8$$

Que 1. Find minimal spanning tree by Kruskal's algorithm.



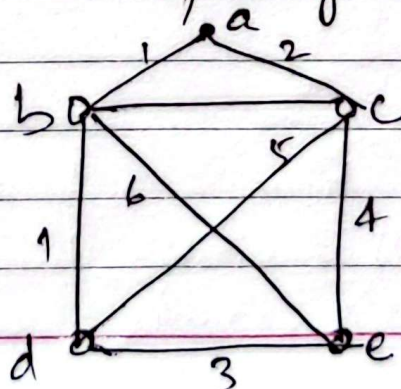
Ans. wt. = 13

Que 2. Find the minimal spanning tree for the graph using both Kruskal's and Prim's method.



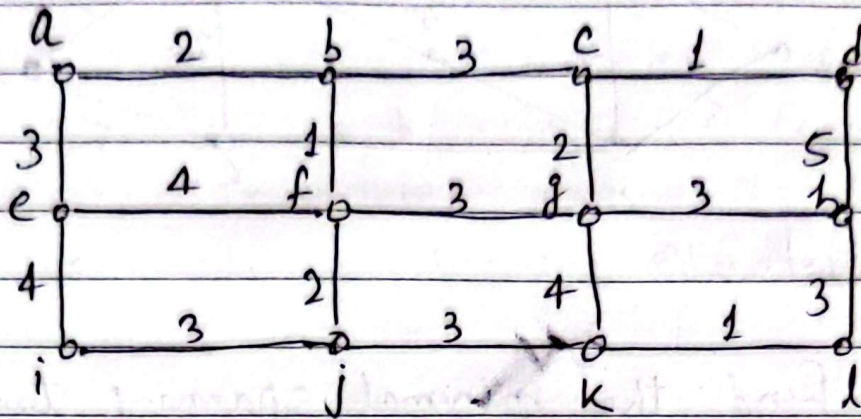
Ans: 12 (weight)

Que 3: Find minimal spanning tree by Prim's method.



~~Notes~~

Que 4 Use (a) Prim's algorithm and (b) Kruskal's algorithm to obtain minimum spanning tree for following graph.



Ans. 24

Network Flows

A network is represented by a weighted graph in which the vertices represent the stations of that network the edges represent the lines which connect the two stations through which the given weight assigned to each edge shows the capacity.

- Capacity is the maximum amount of flow possible per unit of time through that line.

A network is a connected ~~graph~~ digraph N with the following properties.

- (a) There is a unique node (vertex), the source, that has in-degree 0. We generally label the source node 1.
- (b) There is a unique node, the sink (destination), that has out-degree 0. If N has n nodes, we generally label the sink as node n .
- (c) The graph N is labeled. The label, C_{ij} , on each edge (i, j) is a nonnegative number called the wt/capacity of the edge.

For simplicity, we also assume that all edges carry material in one direction only.

Flows

The purpose of a network is to implement a flow of water, oil, electricity, traffic, or whatever the network is designed to carry. Mathematically, a flow in a network N is a function that assigns to each edge (i, j) of N a nonnegative

Notes (flow)

number F_{ij} that does not exceed C_{ij} (capacity).

We also require that for each node other than the source and sink, the sum of the F_{ik} on edges entering node k must be equal to the sum of F_{kj} on edges leaving node k .

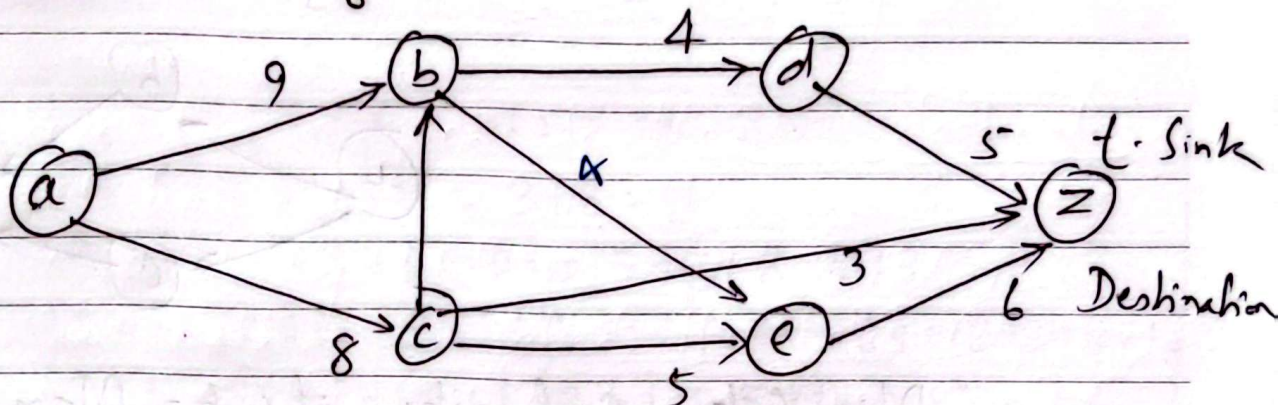
Similarly, the sum of the flows leaving the source must equal the sum of the flows entering the sink. This sum is called the value of the flow, written $\text{value}(F)$. We can represent a flow F by labeling each edge (i, j) with the pair (C_{ij}, F_{ij}) .

Max-Flow Min-Cut Theorem

In a given transport network G , the maximum value of flow i.e., $f_{\max}$ from source to sink is equal to minimum values of the capacities of all the cut in G that separate source from sink.

$$\text{i.e. } f_{\max} = \min C(X, \bar{X})$$

Que. Find maximum flow in the transport network using labeling procedure. Determine the corresponding mincut.



Solⁿ:

Flow:	Path	Flow
1.	$a \rightarrow b \rightarrow d \rightarrow z$	4
2.	$a \rightarrow b \rightarrow e \rightarrow z$	4
3.	$a \rightarrow c \rightarrow e \rightarrow z$	2
4.	$a \rightarrow c \rightarrow z$	3
		13

$\therefore \text{Max. flow} = 13.$

Notes

Cut: Set of edges whose removal divides network into two halves X and Y where

$S \in X$ i.e. Source $\in X$

$T \in Y$ i.e. Sink $\in Y$

$\therefore$ Min cut of above network is

Removal

$b \rightarrow d$

$c \rightarrow z$

$e \rightarrow z$

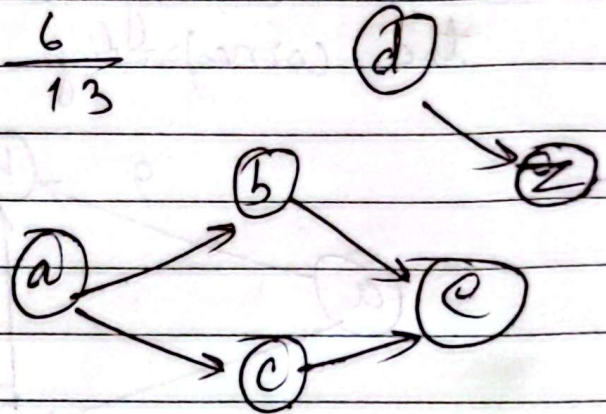
Cut

4

3

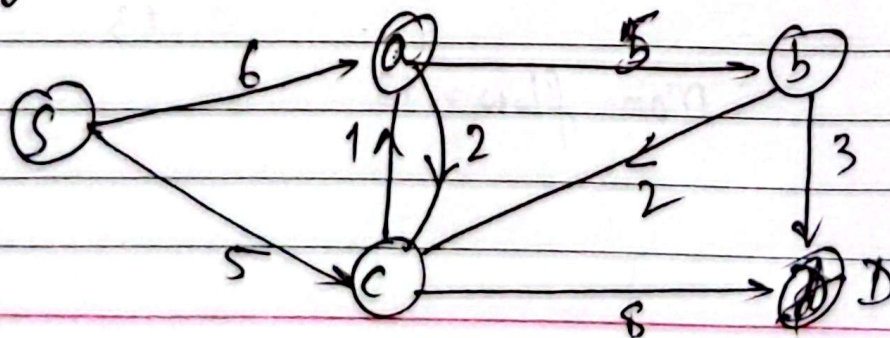
6

13



$\therefore$ Min Cut = Capacity of cut = Max-flow.

Que: In the given transport network, find all cut set that separates source from sink. Also find out the cut set with minimum capacity.



<u>S. No.</u>	<u>X</u>	<u>$\bar{X}$</u>	<u>Cut(X, $\bar{X}$)</u>
1.	{S}	{a, c, b, D}	11
2.	{S, a, c, b}	{D}	11
3.	{S, a, c}	{b, D}	13
4.	{S, c}	{a, b, D}	15
5.	{S, a}	{b, c, D}	12
6.	{S, a, b}	{c, D}	12

There are two cut-set having minimum capacity i.e. 11. They are

- (i) {S} {a, c, b, D}
(ii) {S, a, c, b} {D}

1. {S, a} + {S, c} = 6 + 5 = 11

2. {b, D} + {c, D} = 3 + 8 = 11

3. {a, b} + {b, c} + {c, D} = 5 + 2 + 8 = 13

4. {S, a} + {c, a} + {a, c} + {b, c} + {c, D} = 6 + 1 + 2 + 2 + 8 = 19

5. {S, c} + {c, a} + {a, c} + {a, b} = 5 + 1 + 2 + 5 = 13

6. {S, c} + {c, a} + {a, c} + {b, c} + {b, D}
= 5 + 1 + 2 + 2 + 3 = 13